AGA KHAN UNIVERSITY EXAMINATION BOARD

HIGHER SECONDARY SCHOOL CERTIFICATE

CLASS XII EXAMINATION

APRIL/ MAY 2017

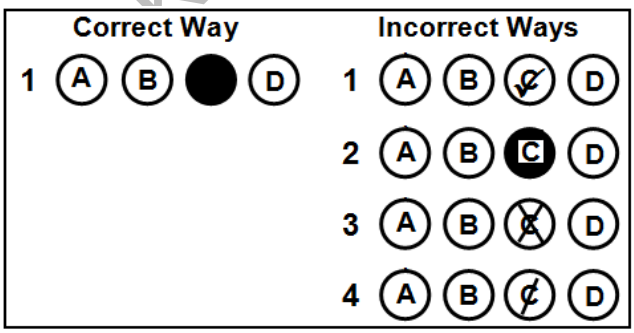
Physics Paper I

Time: 40 minutes Marks: 30

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INSTRUCTIONS

1. Read each question carefully.
2. Answer the questions on the separate answer sheet provided. DO NOT write your answers on the question paper.
3. There are 100 answer numbers on the answer sheet. Use answer numbers 1 to 30 only.
4. In each question there are four choices A, B, C, D. Choose ONE. On the answer grid black out the circle for your choice with a pencil as shown below.



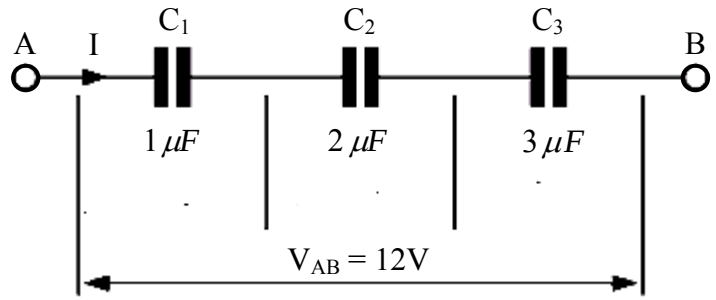
Candidate's Signature

5. If you want to change your answer, ERASE the first answer completely with a rubber, before blacking out a new circle.
6. DO NOT write anything in the answer grid. The computer only records what is in the circles.
7. You may use a scientific calculator if you wish.

1. Three capacitors of different capacitances are connected in series. Which of the following quantities will remain the same across all the capacitors?

   A. Charge
   B. Voltage
   C. Current
   D. Capacitance

2. In the given diagram, three capacitors are connected in series. Their equivalent capacitance will be



   A. 0.545μF
   B. 1.833μF
   C. 2 μF
   D. 6 μF

3. Which of the following will be produced by the displacement of positive and negative charges in an object due to an electric field?

   A. Electric flux
   B. Magnetic flux
   C. Electric dipole
   D. Magnetic intensity

4. If the distance between two identical charges becomes doubled, then the force of repulsion between them decreases by

   A. two-third of the previous value.
   B. one-third of the previous value.
   C. one-fourth of the previous value.
   D. three-fourth of the previous value.

5. If we increase the length of the wire, then its resistance will

   A. increase.
   B. decrease.
   C. become zero.
   D. remain the same.

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6. How much power is dissipated by a resistor of resistance 10 Ω when current of 10 A is passing through it?

   A. 1 watt
   B. 10 watt
   C. 100 watt
   D. 1,000 watt

7. If the terminal potential difference of a battery is equal to its electromotive force (e.m.f.), then the internal resistance of the battery becomes

   A. zero.
   B. infinite.
   C. constant.
   D. maximum.

8. An ohmmeter is used to measure the

   A. current.
   B. voltage.
   C. resistance.
   D. conductance.

9. If an electron is allowed to move perpendicular to the uniform magnetic field, then it will follow the direction in a

   A. zig-zag path.
   B. circular path.
   C. straight path.
   D. triangular path.

10. When the filament in a cathode rays oscilloscope (CRO) is heated, then the emitted particles are

    A. protons.
    B. neutrons.
    C. electrons.
    D. positrons.

11. In an ideal transformer, the turn ratio from primary to secondary is 2:3. If the input voltage is 220 V at the frequency of 50 Hz, then the output voltage and its frequency will be

    A. 110 V and 50 Hz
    B. 110 V and 75 Hz
    C. 330 V and 50 Hz
    D. 330 V and 75 Hz

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12. The amount of energy stored per unit volume inside the solenoid is called

A. flux.
B. power.
C. energy density.
D. inductor energy.

13. Lenz's law helps to understand the law of conservation of

A. mass.
B. matter.
C. energy.
D. momentum.

14. The combined effect of resistance and reactance in a circuit is known as

A. inductor.
B. capacitor.
C. impedance.
D. modulation.

15. Which of the following options correctly describes the amplitude modulation (AM) and frequency modulation (FM) radiowaves?

	Amplitude Modulation (AM)	**Frequency Modulation (FM)**
A	Longer range and able to travel around obstacles	Shorter range and unable to travel around obstacles
B	Shorter range and unable to travel around obstacles	Longer range and able to travel around obstacles
C	Longer range but unable to travel around obstacles	Shorter range but able to travel around obstacles
D	Affected more by electrical interference	Affected less by electrical interference

16. Which of the following statements is CORRECT about the phase lagging in a capacitive circuit?

A. Current leads the voltage.
B. Voltage leads the current.
C. Voltage leads the resistance.
D. Resistance leads the current.

17. The ratio of applied stress to the VOLUMETRIC strain is called

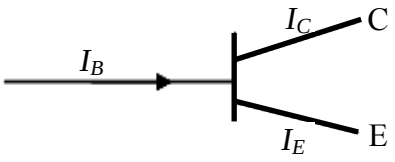
A. Bulk modulus.
B. Young's modulus.
C. modulus of rigidity.
D. modulus of elasticity.

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18. A dopped semi-conductor is called

A. ceramic.
B. intrinsic.
C. extrinsic.
D. insulator.

19. The current gain by the given transistor in the common-base circuit is defined by the ratio



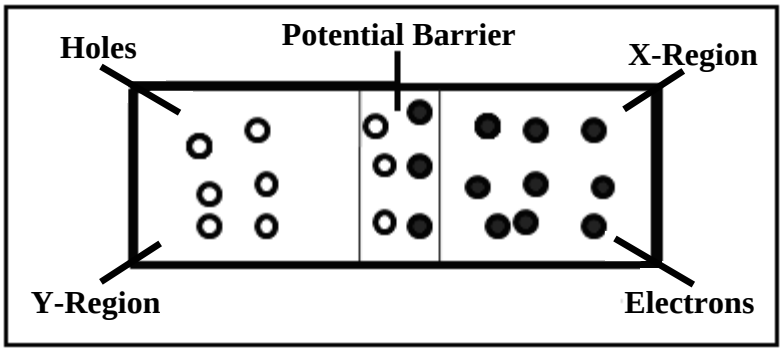
A. $\frac{I_C}{I_E}$

B. $\frac{I_C}{I_B}$

C. $\frac{I_E}{I_B}$

D. $\frac{I_E}{I_C}$

20. For every electron that diffuses across a junction and combines with a hole, a positive charge is left in the **X** region and a negative charge is created in the **Y** region forming a potential barrier.



The **X** and **Y** regions are

	X-Region	Y-Region
A	*n*	*p*
B	*p*	*n*
C	*p*	*p*
D	*n*	*n*

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21. Which of the following uses a reverse biased diode?

A. Rectifier
B. Calculator
C. Photodiode
D. Light emitting diode

22. When a black body emits radiations then these radiation are

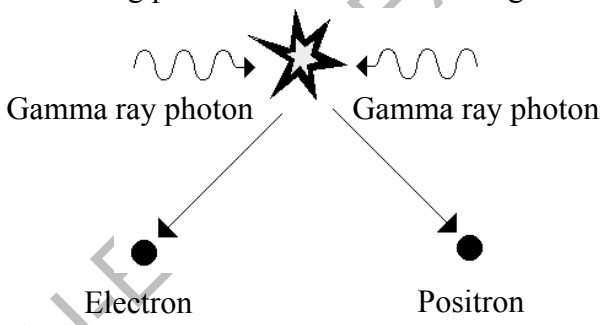
A. independent of the size of the black body.
B. dependent upon the size of the black body.
C. independent of the temperature of the radiator.
D. dependent upon the temperature of the radiator.

23. The maximum energy carried by the photoelectrons depend upon the

I. area of the metal surface.
II. intensity of incident light.
III. energy of incident photon.
IV. frequency of incident light.

A. I and II
B. II and III
C. II and IV
D. III and IV

24. Which of the following phenomena is shown in the given diagram?



A. Annihilation
B. Pair production
C. Compton's effect
D. Photoelectric effect

25. According to Bohr's atomic model, the energy of the farthest shell from the nucleus is

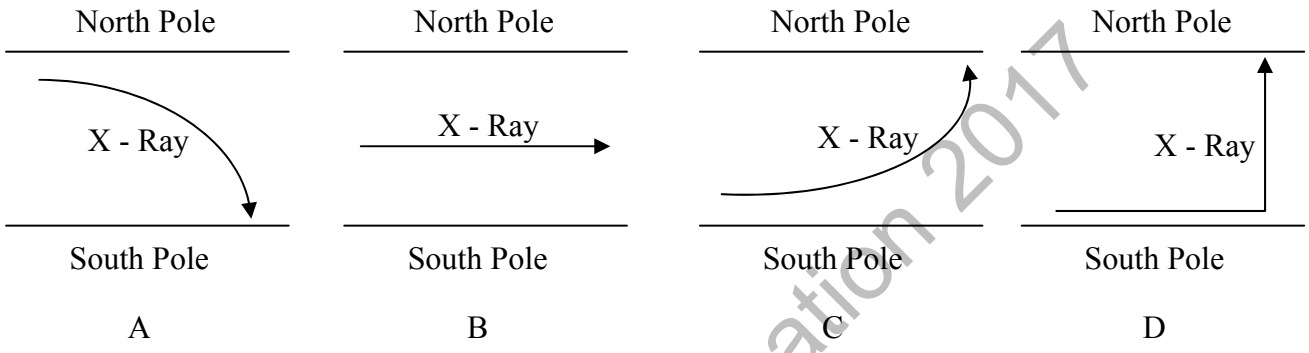
A. zero.
B. infinite.
C. minimum.
D. maximum.

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26. All of the following electromagnetic radiations have wavelength longer than X-rays EXCEPT

A. visible rays.
B. gamma rays.
C. infrared rays.
D. ultraviolet rays.

27. Which of the following options shows the behaviour of X-rays when it passes through magnetic field?



28. The emission of one alpha particle from the nucleus of an atom produces a change of

A. – 4 in atomic mass.
B. – 4 in atomic number.
C. – 1 in atomic mass.
D. – 1 in atomic number.

29. Geiger Muller counter is used to detect

A. energy.
B. particle.
C. electric charge.
D. ionising radiation.

30. According to the modern view of building blocks of matter, a proton is assumed to be made up of

A. one quark.
B. two quarks.
C. three quarks.
D. four quarks.

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END OF PAPER

Please use this page for rough work

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